L4

CLASS X - SCIENCE

CONTROL AND COORDINATION

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How does the Nervous Tissue cause Action?

- Nervous tissue **collects information, sends it around the body, processes it, makes decisions, and conveys those decisions to muscles for action.**
- When the actual action or movement is to be performed, muscle tissue does the final job.

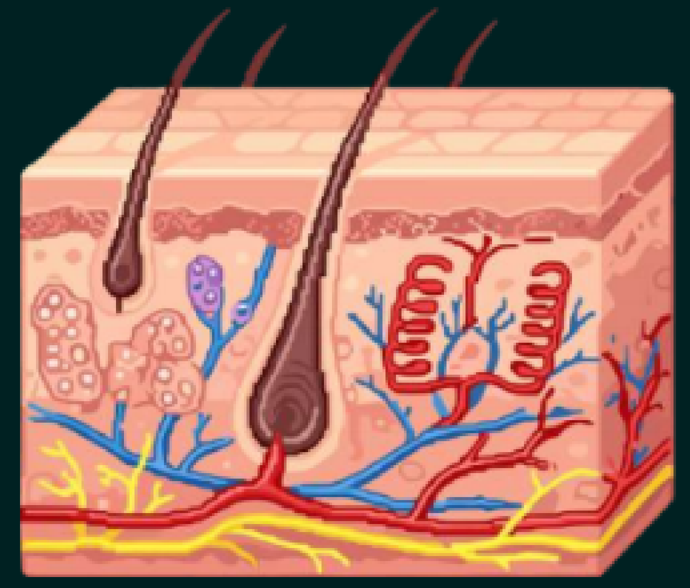
How Muscles Move

- When a nerve impulse reaches the muscle, the muscle fibre must move.
- At the cellular level, muscle cells move by changing their shape so that they shorten (contract).

How Muscle Cells Change Shape (Chemistry behind it)

- Muscle cells have **special proteins.**
- These proteins change both their shape and their arrangement in the cell in response to nervous electrical impulses.
- This new arrangement of proteins gives the muscle cells a shorter form → causing contraction/movement.

Glands



Glands are special organs in our body that make and release substances like **hormones, enzymes, sweat, or saliva.**

Exocrine glands

- *Release their secretions through ducts to a surface or organ.*
- *Example: Sweat glands, Salivary glands, Tear glands*

Endocrine glands

- *Release hormones directly into the blood.*
- *Example: Pituitary gland, Thyroid gland, Adrenal gland*

Exocrine Glands	Endocrine Glands
Have **ducts** → pour secretions to a specific organ or surface.	**Ductless glands** → release hormones directly into the blood.
Secrete substances like **enzymes, saliva, sweat, mucus, etc.**	Secrete **hormones** only.
Act **locally** (limited effect where ducts open).	Act on **distant target organs/tissues** through blood.
Examples: **Salivary glands, Sweat glands, Tear glands.**	Examples: **Pituitary gland (growth hormone), Thyroid gland (thyroxin), Adrenal gland (adrenaline), Pancreas (insulin), Testis (testosterone), Ovary (oestrogen).**

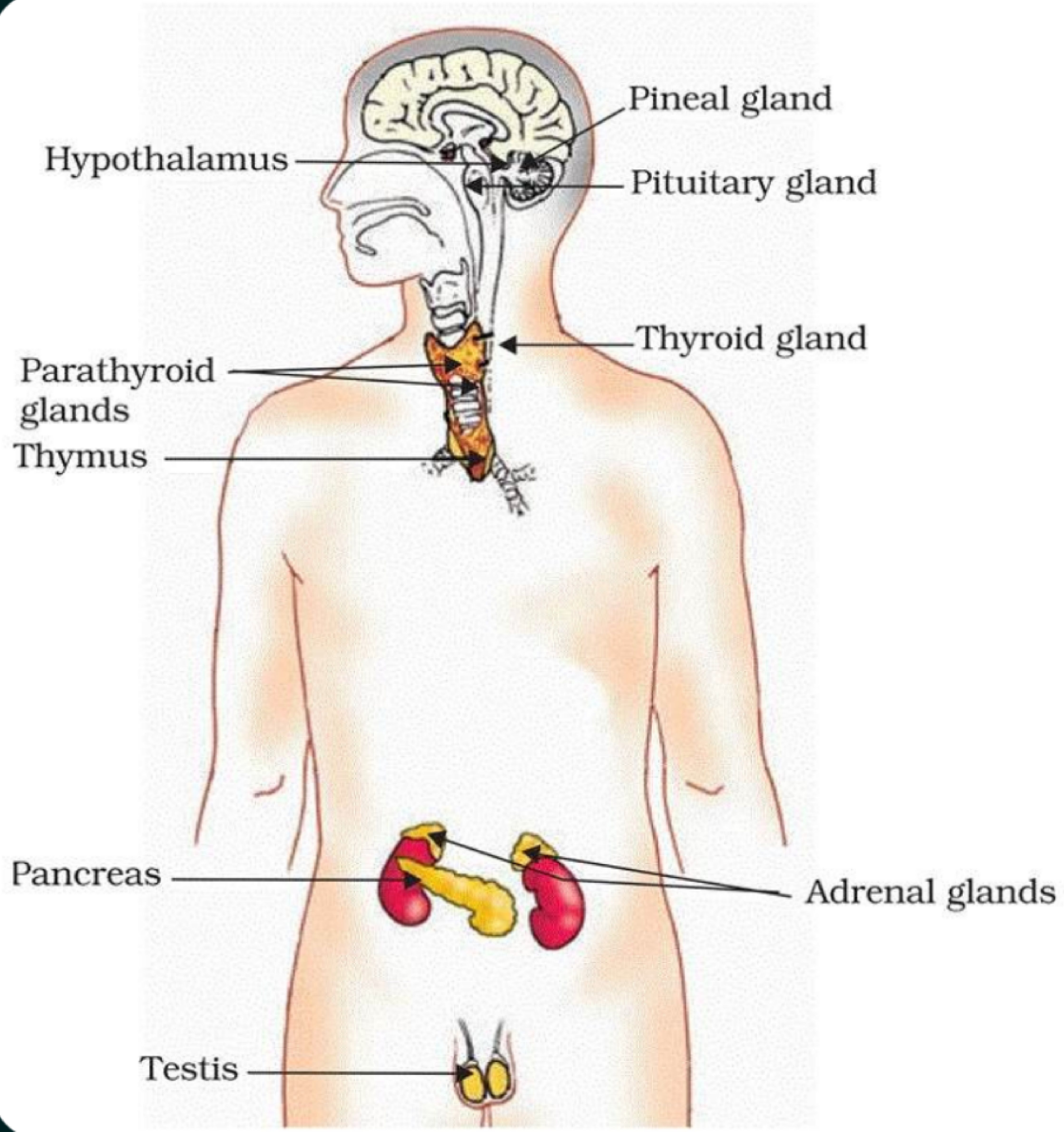
Q. Why is chemical communication better than electrical impulses as a means of communication between cells in multicellular organisms?

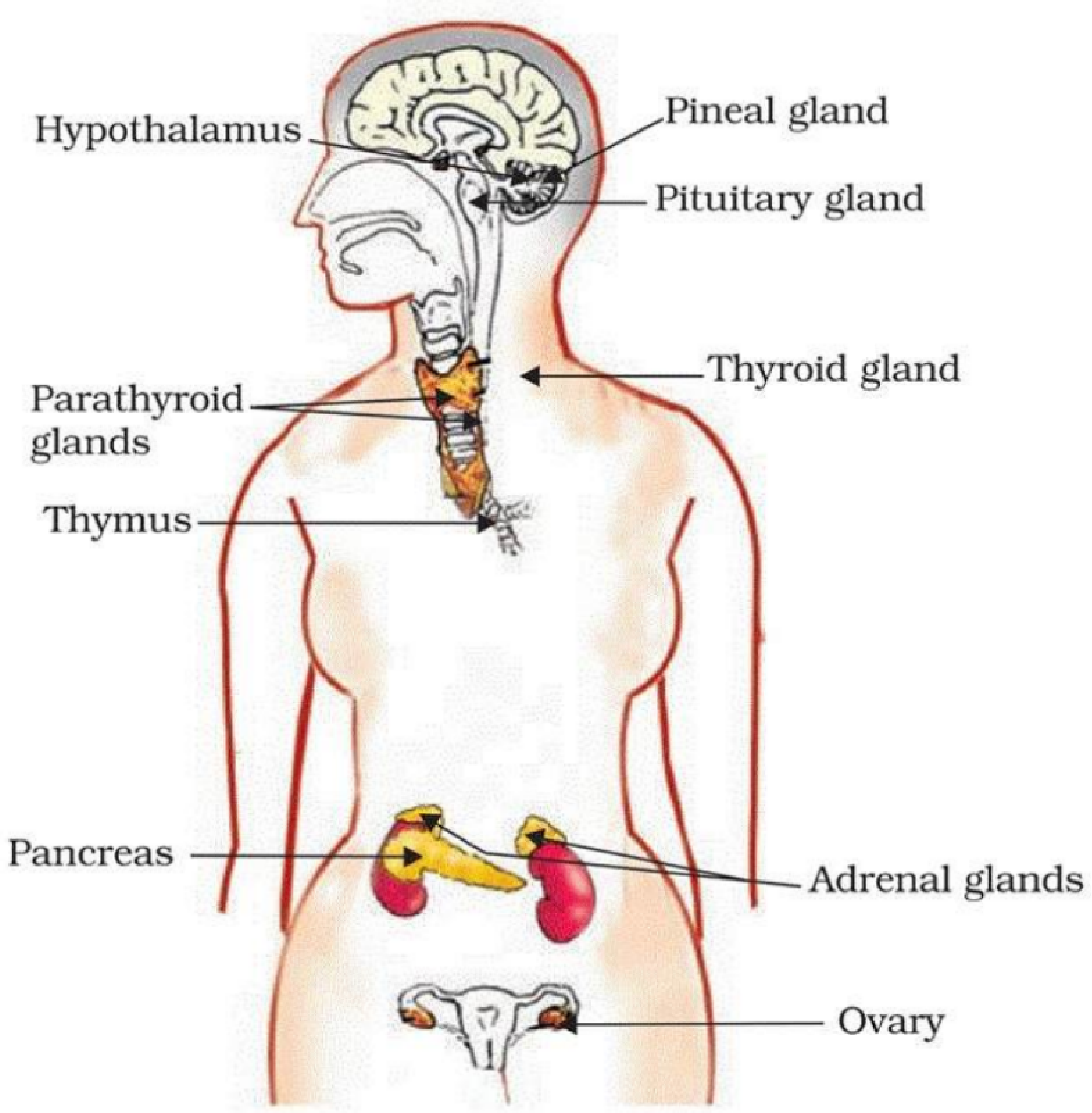
Ans. Electrical impulses have a limited range → they can only pass through connected neurons, not reach every cell of the body.

- *Also, neurons need a recovery time before transmitting the next impulse.*

Chemical communication (through hormones) is better because:

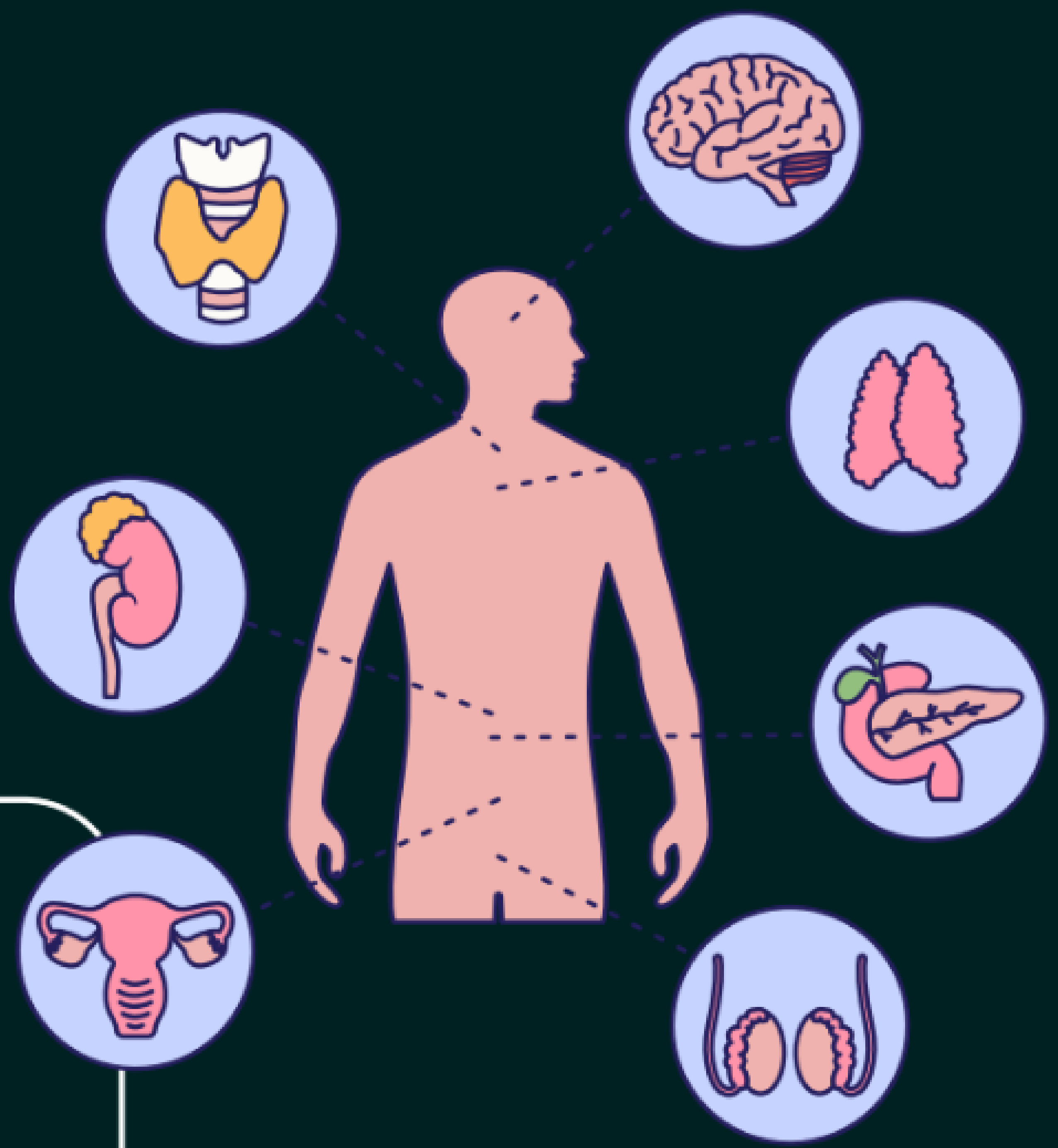
- *Wider range → Hormones can travel through blood and reach all parts of the body.*
- *Long-lasting effect → Chemical signals act for longer duration, unlike the short-lived electrical impulses.*
- *Target-specific → Hormones act only on specific target tissues or organs.*


Pineal gland
Hypothalamus
Pituitary gland
Thyroid gland
Parathyroid glands
Thymus
Pancreas
Adrenal glands
Testis


Pineal gland
Hypothalamus
Pituitary gland
Thyroid gland
Parathyroid glands
Thymus
Pancreas
Adrenal glands
Ovary

Hormone

Hormones are chemical messengers secreted by endocrine glands directly into the blood. They are carried to specific target organs or tissues, where they control and coordinate various activities of the body.



Q. Why Are Hormones Needed?

Electrical impulses transmitted by nerve cells:

- Can reach only those cells that are connected by nervous tissue.
- Produce rapid but short-lived responses.

Hormones:

- Are transported through blood.
- Can reach different parts of the body.
- Produce widespread and longer-lasting changes.

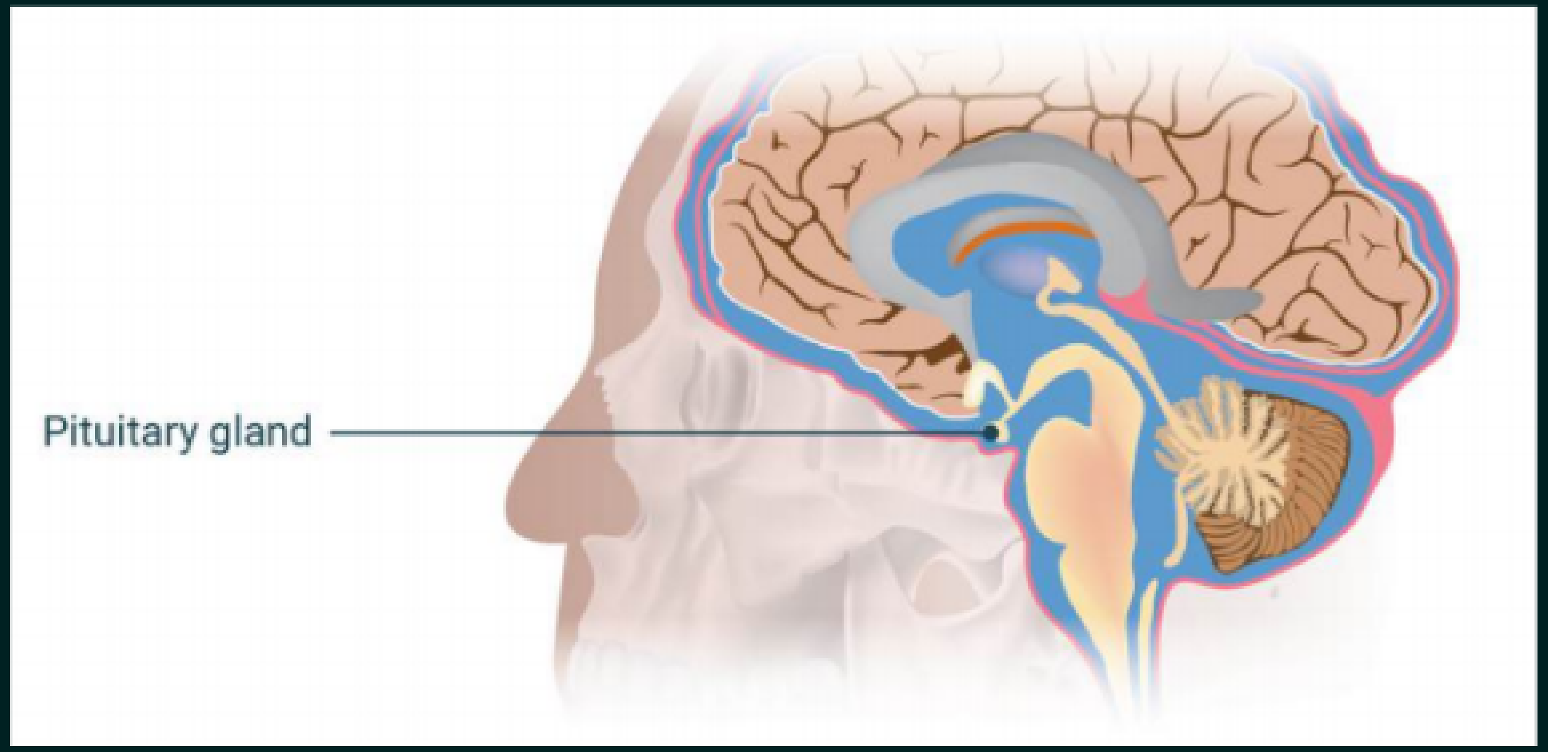
Pituitary gland

- **Hormone: Growth hormone**

- Pituitary gland is *present just below the brain.*

Function:

- Regulates the growth and development of the body.
- Stimulates growth in different organs.

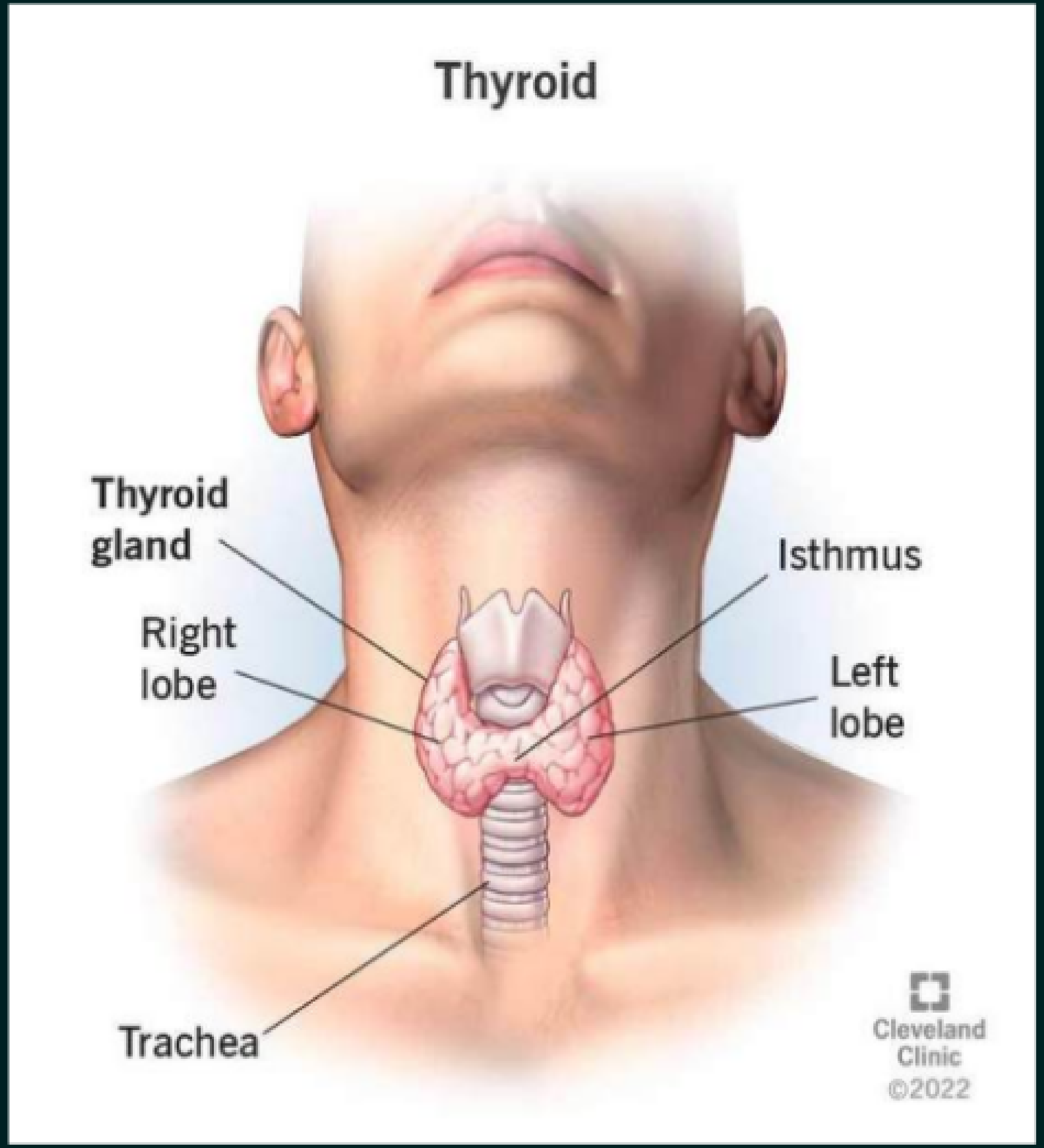


Dwarfism - A person having a deficiency of growth hormone in childhood remains very short and becomes a dwarf.

Gigantism - a person with an excess of growth hormone becomes very tall (a condition known as gigantism).

Thyroid gland

- **Hormone: Thyroxin**
- **Requirement for its production: Iodine**



Functions

- Thyroxin regulates the metabolism of:
  - Carbohydrates
  - Proteins
  - Fats
- It maintains the proper balance of metabolism required for normal growth.

Iodine Deficiency

- Iodine is necessary for the synthesis of thyroxin.
- Deficiency of iodine may cause **goitre.**
- A major symptom of goitre is a swollen neck due to enlargement of the thyroid gland.
- Therefore, we should use iodised salt in our diet.

Source of Iodine

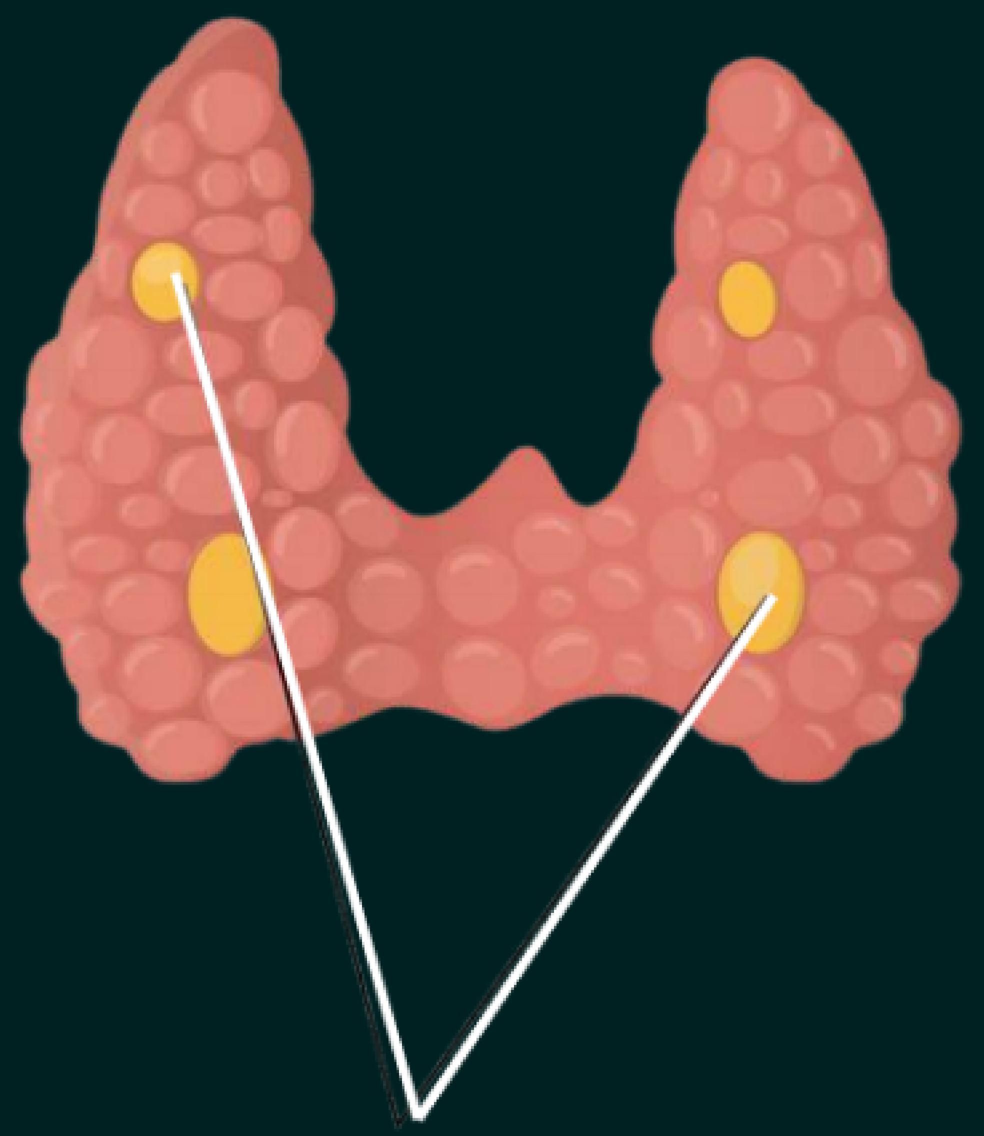
Seaweed, Fish, shellfish, Table salts labeled “iodized”, Dairy (milk, cheese, yogurt), Eggs, Beef liver, Chicken.

Parathyroid gland

- **Hormone: Parathormone**

- The parathyroid glands are four small glands located within the thyroid gland.

- They produce a hormone called **parathormone**, which helps regulate the levels of calcium and phosphate in the blood, ensuring proper functioning of nerves and muscles.

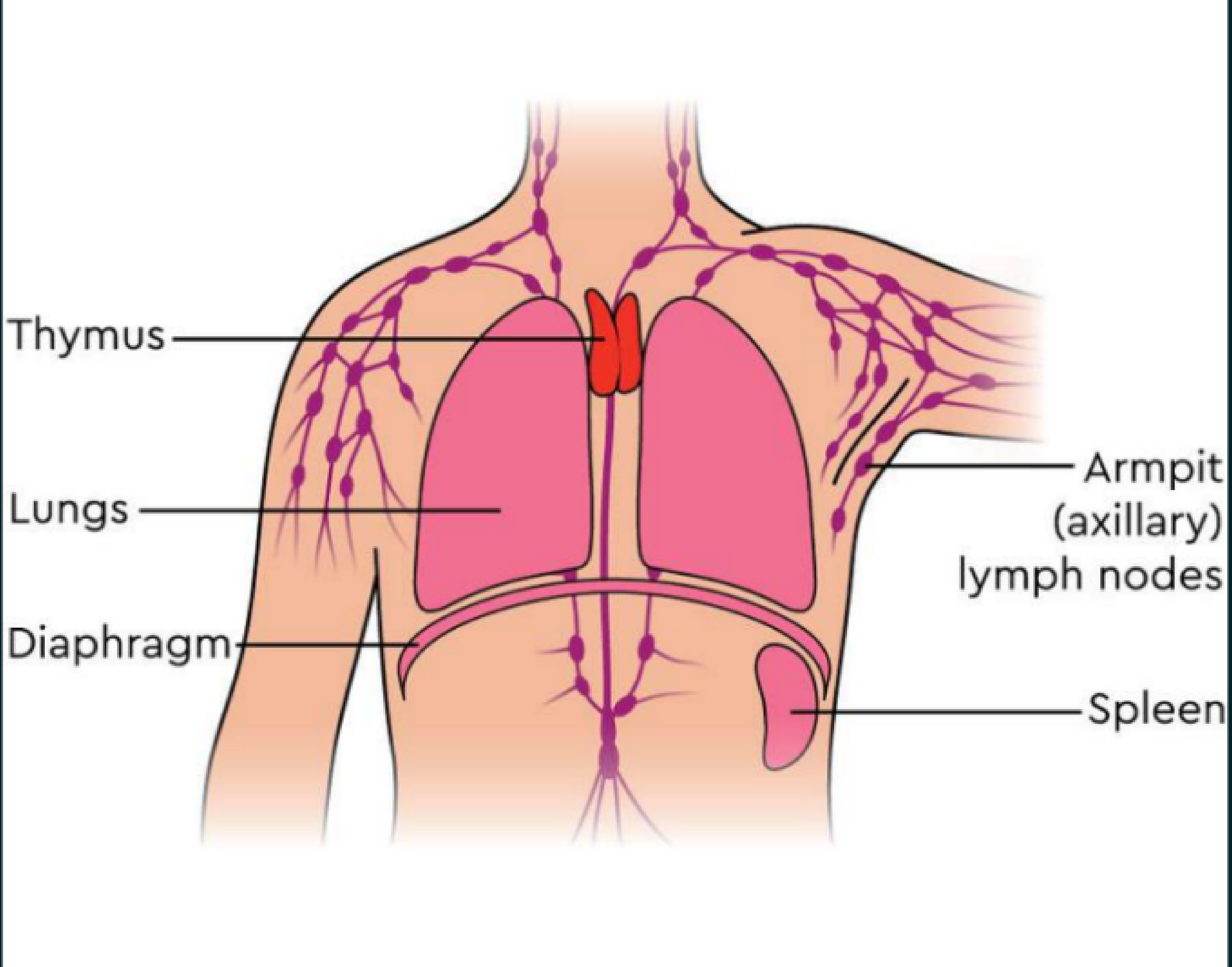


Parathyroid gland

Thymus gland

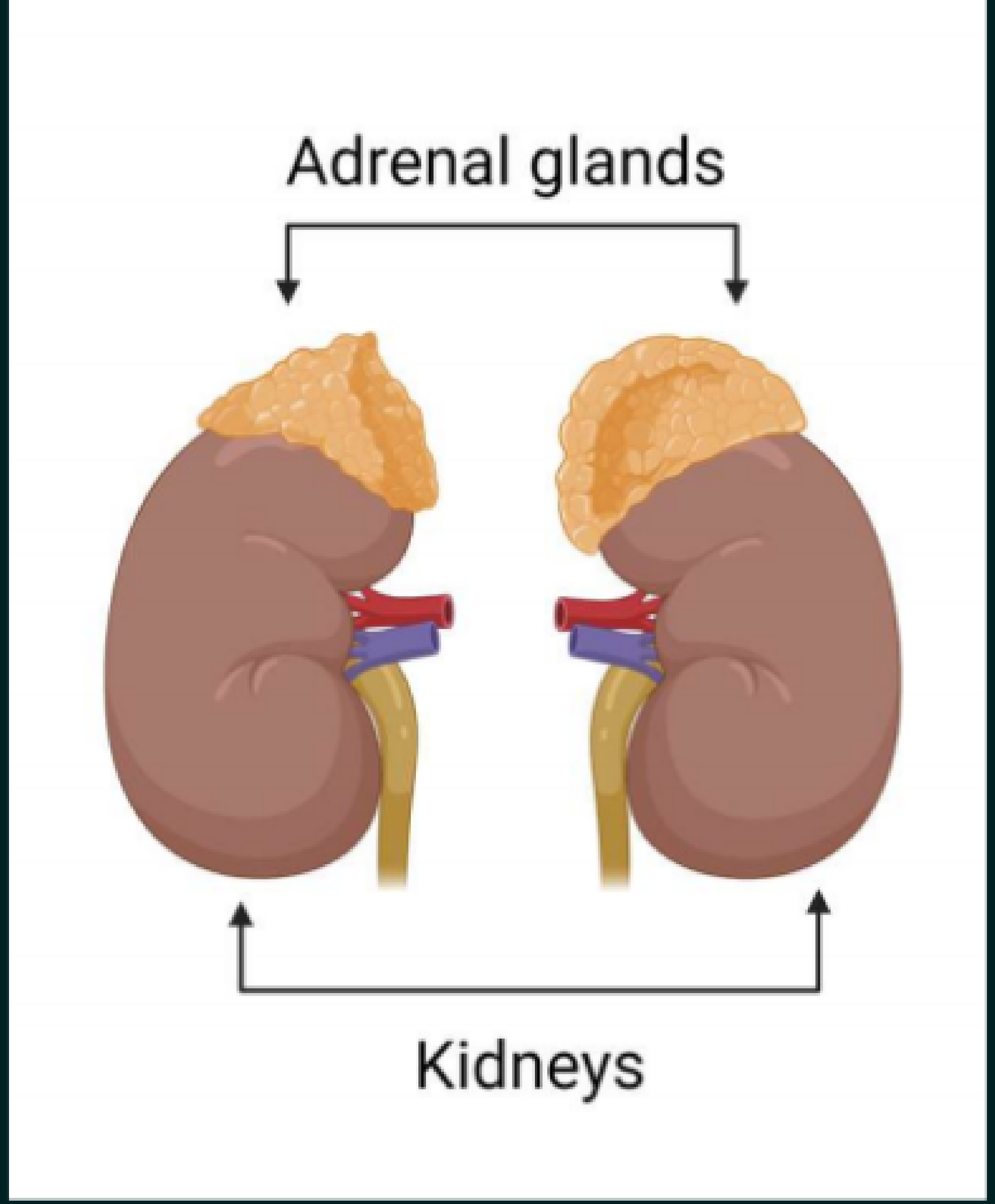
- **Hormone: Thymus hormone**

- The thymus gland is located in the lower neck and upper chest region. It secretes the thymus hormone, which is important for the development of the body's immune system.
- The gland is large in children but gradually shrinks after puberty.



Adrenal glands

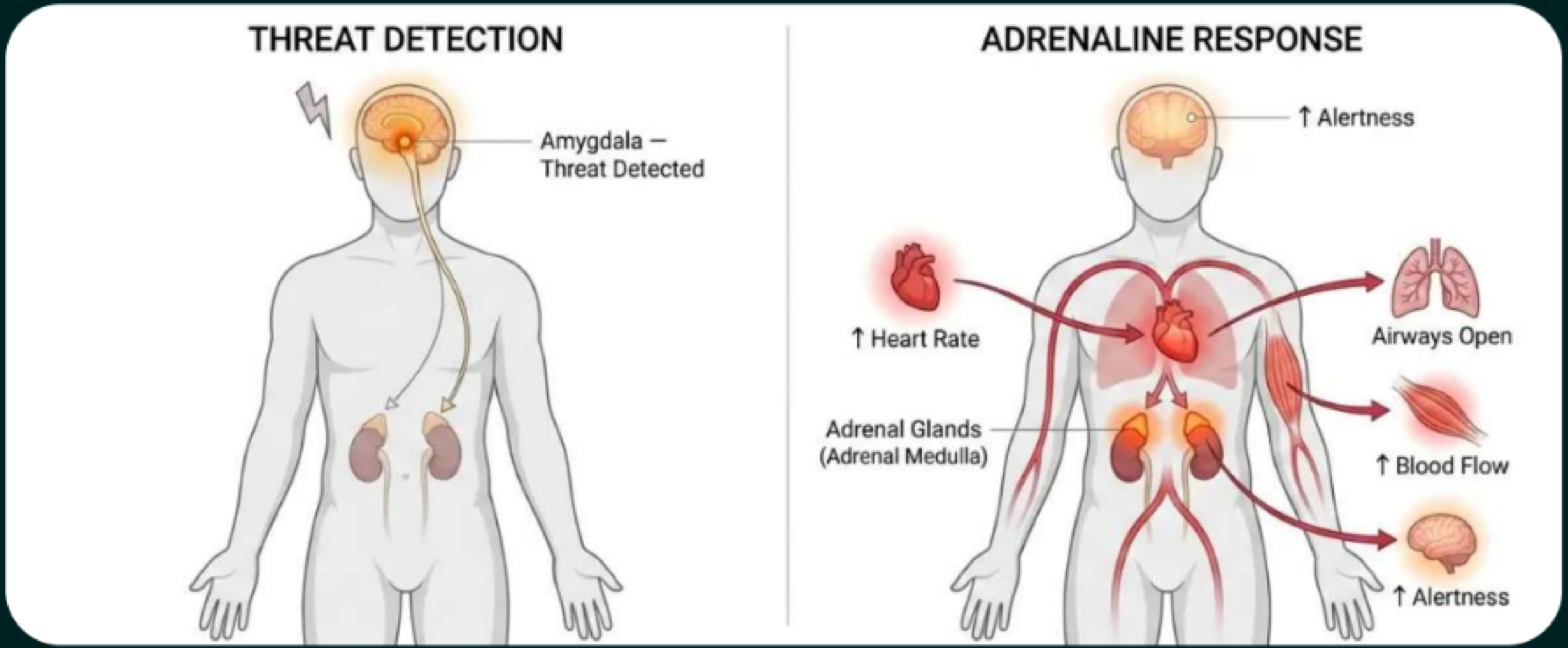
- Hormone: Adrenaline
- Location: Above the kidneys
- Released during: Fear, stress, danger or emergency



Adrenaline prepares the body for fighting or running away. This is called the **fight-or-flight response.**

Effects of Adrenaline:

- Heart beats faster.
- More oxygen is supplied to the muscles.
- Breathing rate increases due to contractions of the diaphragm and rib muscles.
- Blood supply to the digestive system and skin is reduced.
- More blood is diverted towards the skeletal muscles.
- The body becomes ready to face an emergency situation.

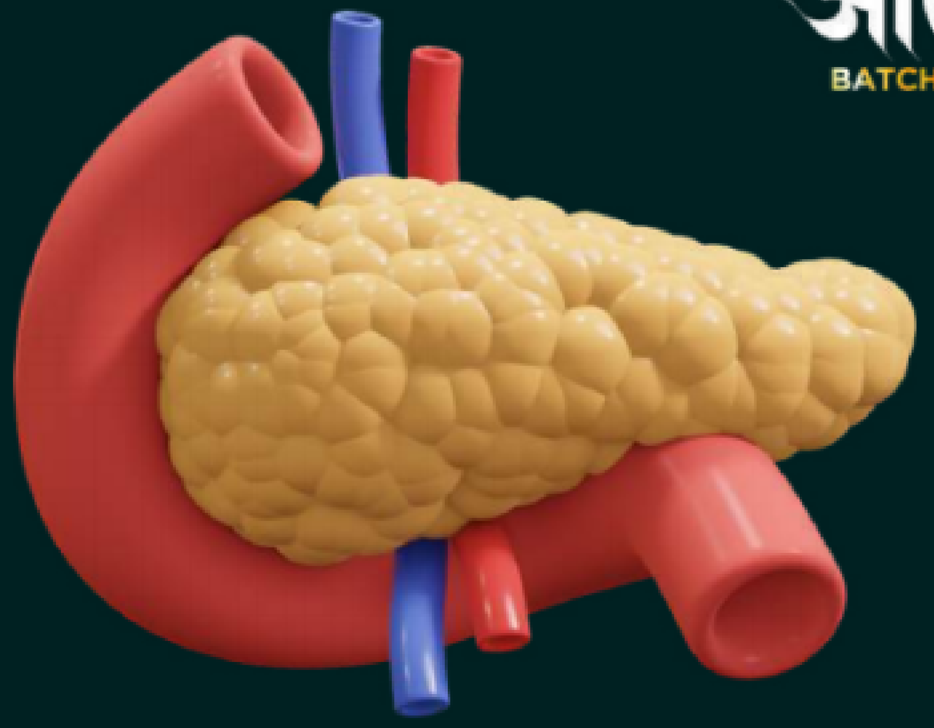


Flowchart

Danger or fear → Adrenal glands release adrenaline → Heartbeat and breathing rate increase → More blood and oxygen reach muscles → Body becomes ready for action

Pancreas

- **Hormone: Insulin**



The pancreas is called a **dual or mixed gland** because it performs both endocrine and exocrine functions.

Endocrine Function: The endocrine part of the pancreas secretes hormones directly into the bloodstream.

- It secretes insulin, which helps regulate the blood sugar level.
- If insulin is not secreted in the required amount, the blood sugar level rises.
- This condition may lead to diabetes.
- Some diabetic patients are advised to take insulin injections.

Exocrine Function: The exocrine part of the pancreas secretes digestive enzymes through a duct into the small intestine. These enzymes help in the digestion of carbohydrates, proteins and fats.

- Insulin helps in regulating the blood sugar level.
- If insulin is not secreted in the proper amount, the **sugar level in the blood rises.**
- Increased blood sugar may cause several harmful effects and lead to **diabetes.**
- Some diabetic patients are advised to take insulin injections.

Feedback Regulation of Insulin:

- When the blood sugar level rises, the cells of the pancreas detect it.
- The pancreas responds by producing more insulin.
- As the blood sugar level falls, insulin secretion is reduced.

Blood sugar rises → Pancreas releases more insulin → Blood sugar falls → Insulin secretion decreases

Testes (Male)

- **Hormone: Testosterone**

Responsible for the development of **male sex organs and male secondary sexual characteristics.**

Examples:

- Growth of facial hair
- Deepening of voice
- Development of male reproductive organs

- These changes occur during male puberty, which typically begins around the age of 13 to 14 years.

Ovaries (Female)

Pregnancy

- **Hormone: Oestrogen** + Progesterone

Responsible for the development of **female sex organs and female secondary sexual characteristics.**

Helps regulate the menstrual cycle.

Examples:

- Development of breasts
- Widening of hips
- Development of female reproductive organs

All the changes caused by estrogen occur at an age of 10 to 12 years.

Role of the Hypothalamus

Brain ?

- The hypothalamus controls the **release of many hormones.**
- It produces releasing hormones that stimulate the pituitary gland.

Example:
When the growth hormone level is low:

Hypothalamus releases growth hormone-releasing factor → Pituitary gland is stimulated → Pituitary releases growth hormone

Hormone	Secreted by	Main functions	Deficiency/Disorder
Adrenaline	Adrenal glands	Prepares the body for a **fight-or-flight response** during frightening or stressful situations. It increases heartbeat and breathing rate and diverts more blood towards skeletal muscles.	
Thyroxin	Thyroid gland	Regulates the metabolism of carbohydrates, proteins and fats to maintain the proper balance required for growth. Iodine is necessary for its synthesis.	Iodine deficiency may cause **goitre**, characterised by a swollen neck.
Growth hormone	Pituitary gland	Regulates the growth and development of the body.	Its deficiency during childhood causes **dwarfism**.
Insulin	Pancreas	Regulates the blood sugar level.	If it is not secreted in the proper amount, the blood sugar level rises and may lead to **diabetes and other harmful effects**.
Testosterone	Testes in males	Causes changes associated with puberty in males and supports the development of male sex organs and secondary sexual characteristics.	
Oestrogen	Ovaries in females	Causes changes associated with puberty in females, supports the development of female sex organs and regulates the menstrual cycle.	

Feedback Mechanism

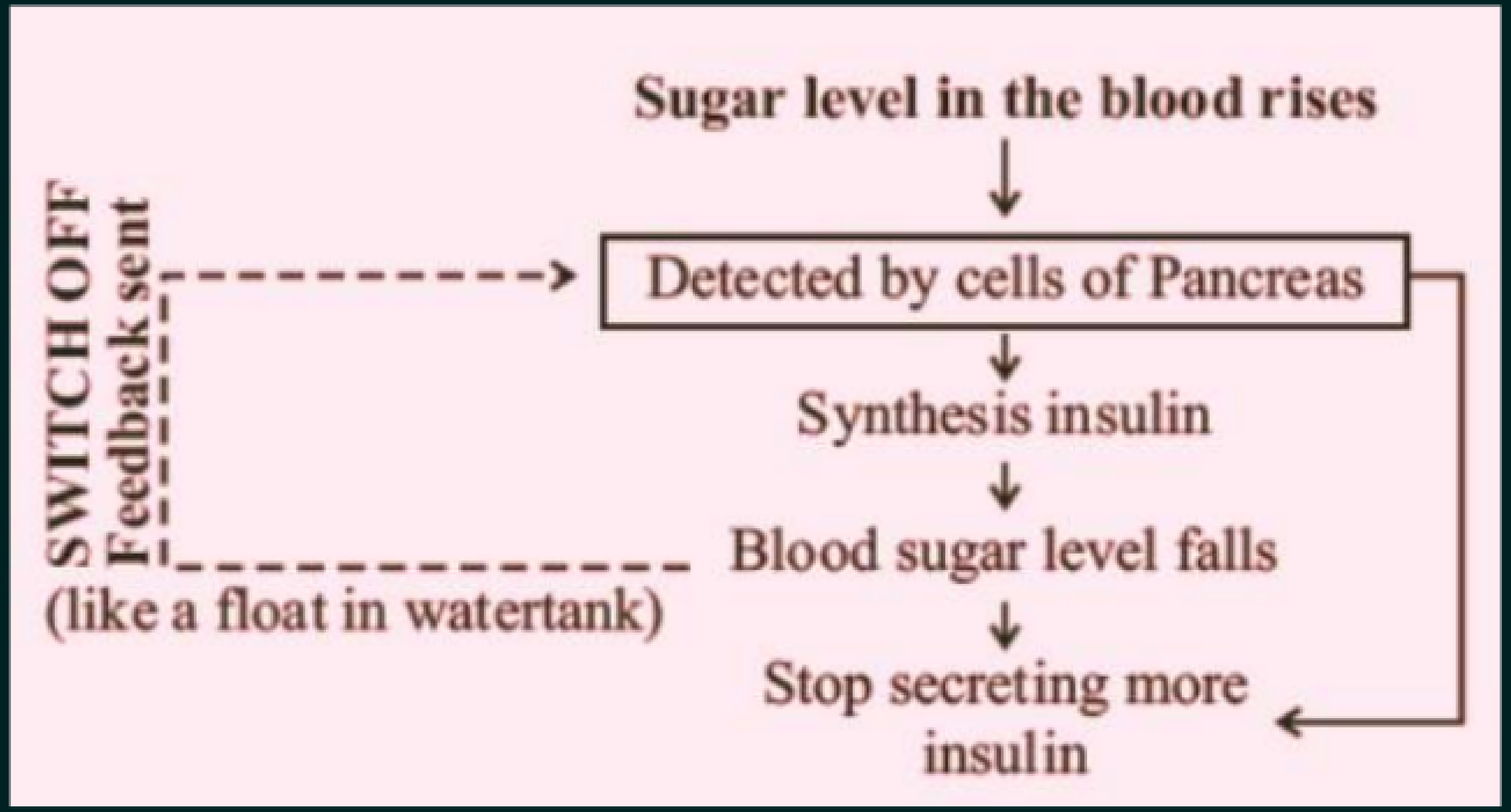
Hormones must be released in precise quantities. Their timing and amount are controlled through a feedback mechanism.

Regulation of Blood Sugar

- When blood sugar level rises, cells of the pancreas detect it.
- The pancreas releases more insulin.
- Insulin lowers the blood sugar level.
- When blood sugar level falls, insulin secretion is reduced.

Flowchart

- Blood sugar rises → Pancreas detects the change → More insulin is released → Blood sugar falls → Insulin secretion decreases
- This mechanism maintains the correct amount of hormone in the body.



LETS'S TEST YOUR MEMORY

Q. Given below are four statements regarding endocrine glands and hormones. Select the correct statements:

(I) Hormones are secreted by endocrine glands.

(II) Some endocrine glands secrete hormones in response to other hormones.

(III) Hormones are required in large quantities.

(IV) Many hormones are controlled by some form of feedback mechanism.

Options:

(a) Both (I) and (II)

(b) Both (I) and (III)

(c) (I), (II) and (III)

(d) (I), (II) and (IV)

LETS'S TEST YOUR MEMORY

Q. Thyroxin regulates the metabolism of:

(a) Only carbohydrates

(b) Only proteins

(c) Only fats

(d) Carbohydrates, proteins and fats

LETS'S TEST YOUR MEMORY

Q. Which hormone helps regulate blood sugar levels?

(a) Adrenaline

(b) Insulin

(c) Thyroxin

(d) Testosterone

LETS'S TEST YOUR MEMORY

Q. Which of the following correctly matches the hormone with its gland?

(a) Insulin — Thyroid gland

(b) Adrenaline — Adrenal gland

(c) Thyroxin — Pancreas

(d) Growth hormone — Ovaries

LETS'S TEST YOUR MEMORY

Q. How does our body respond when adrenaline is secreted into the bloodstream?

When adrenaline is secreted into the bloodstream:

- The heart beats faster, increasing the supply of oxygen to the muscles.
- Blood supply to the digestive system and skin is reduced.
- More blood is diverted towards the skeletal muscles.
- The breathing rate increases due to contractions of the diaphragm and rib muscles.

These changes prepare the body to face an emergency situation through the fight-or-flight response.

LETS'S TEST YOUR MEMORY

Q. In animals, hormones can be secreted by one organ and can act on multiple organs. Justify this statement by explaining the effect of a single animal hormone on three organs. [CBSE Practice Set-1, 2023]

Adrenaline, secreted by the adrenal glands, is carried through the blood and acts on different organs:

1. Heart: It increases the heartbeat, supplying more oxygen to the muscles.
2. Digestive system and skin: It reduces their blood supply by contracting the muscles around small arteries, diverting blood towards the skeletal muscles.
3. Diaphragm and rib muscles: It increases their contraction, thereby increasing the breathing rate.

Thus, a single hormone can act on multiple organs and coordinate the body's fight-or-flight response

LETS'S TEST YOUR MEMORY

Q. A hormone 'X' is secreted into the blood when a person faces a frightening or emergency situation.

(a) Identify hormone 'X' and name the gland that secretes it.

(b) Explain its role in dealing with frightening or emergency situations.

(a) Hormone 'X' is adrenaline, secreted by the adrenal glands.

(b) Adrenaline prepares the body for the fight-or-flight response by:

- Increasing the heartbeat and breathing rate
- Increasing blood supply to the muscles
- Increasing the blood glucose level to provide more energy

These changes help the person respond quickly during an emergency.

LETS'S TEST YOUR MEMORY

Q. A doctor has advised Sameer to reduce sugar intake in his diet and do regular exercise after checking his blood test reports.

1. **Which disease do you think Sameer is suffering from?**
2. **Name the hormone responsible for this disease and the organ producing the hormone.**

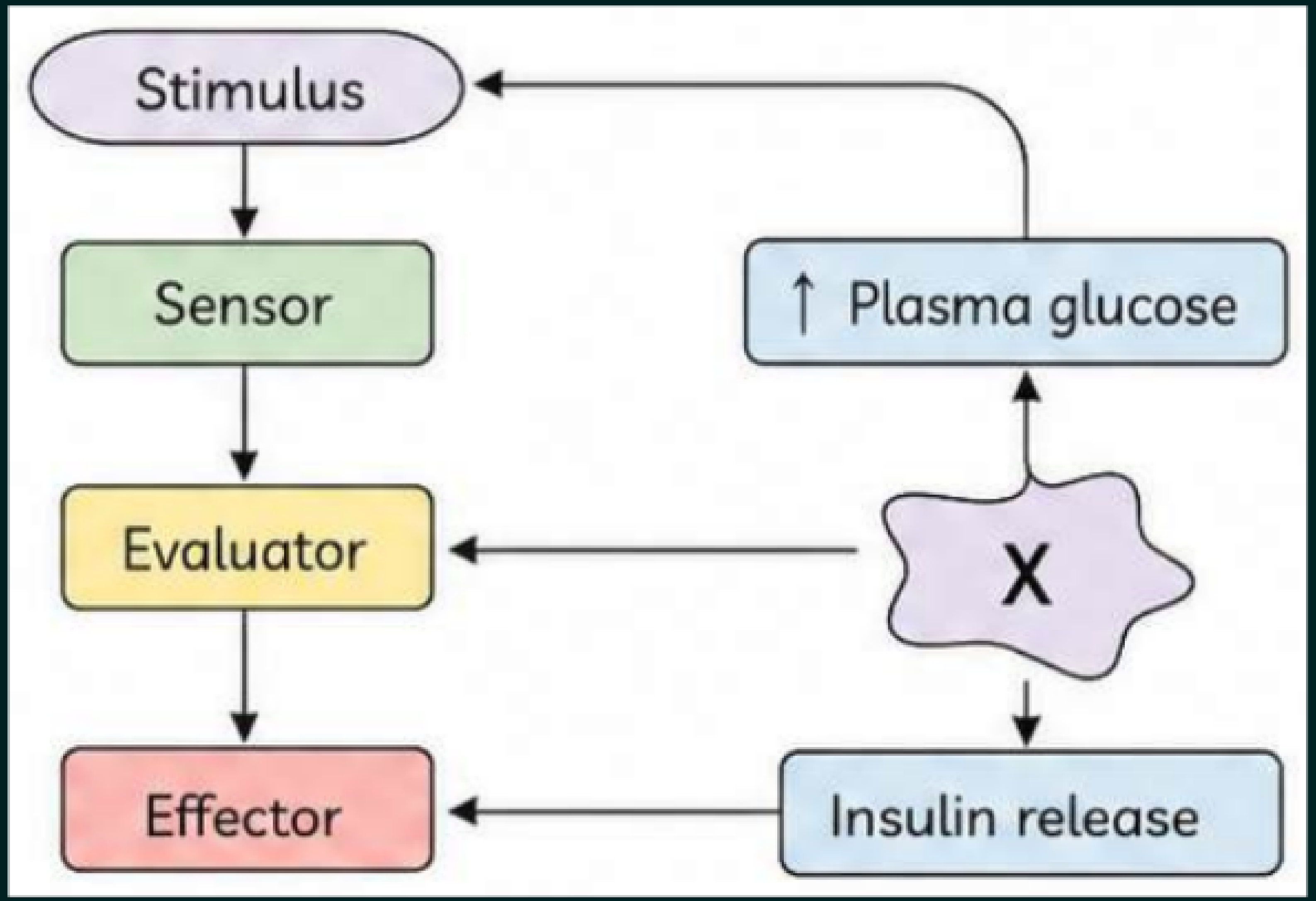
Sameer is most likely suffering from diabetes.

- Hormone responsible: Insulin
- Organ producing it: Pancreas

Insufficient secretion of insulin causes the blood sugar level to rise.

LETS'S TEST YOUR MEMORY

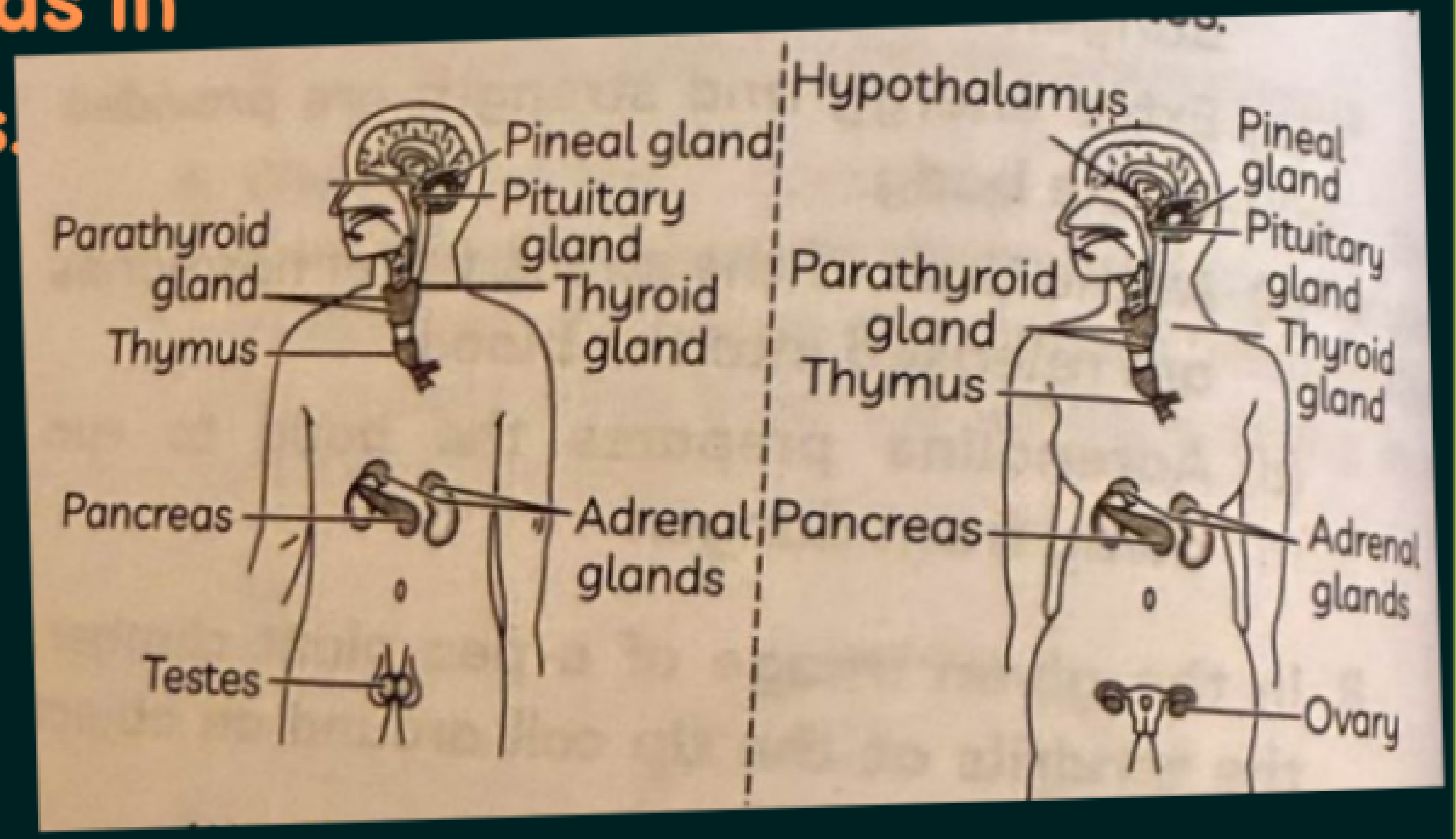
Q. In human beings, hormonal action is largely controlled by a mechanism where the secretion of one hormone is regulated by the action of another. An example of blood glucose level control is shown in the diagram below.
(A) What is the mechanism of hormone action known as?
(B) Which is the sensor X that helps in detecting blood glucose level? What would happen if such a mechanism is absent in humans?



(A) The mechanism is called a feedback mechanism.
(B) X represents the cells of the pancreas, which detect changes in the blood glucose level.
When the blood glucose level rises, the pancreas secretes more insulin. As the blood glucose level falls, insulin secretion is reduced.
If this feedback mechanism were absent, insulin would not be released in the correct amount. The blood sugar level would become uncontrolled, leading to harmful effects such as diabetes.
Increased blood glucose → Pancreas detects it → More insulin is released → Blood glucose decreases → Insulin secretion is reduced

LETS'S TEST YOUR MEMORY

Q. Renu was interested in knowing about the various endocrine glands in the human body, the hormones secreted by them and their functions. The given figure shows the endocrine glands in human males and females.



(A) Which hormone is produced from the pancreas that helps in regulating blood sugar levels?

(B) Which endocrine gland is known as the master gland in human beings?

(C) Which gland secretes the growth hormone?

(A) The pancreas produces insulin, which regulates blood sugar levels.

(B) The pituitary gland is known as the master gland.

(C) The pituitary gland secretes the growth hormone.

LETS’S TEST YOUR MEMORY

Q. The thyroid hormone regulates metabolism in the body. The thyroid gland primarily releases two hormones: Thyroxine (T_4) and Triiodothyronine (T_3), collectively known as thyroid hormones. The thyroid-stimulating hormone (TSH), produced by the pituitary gland, regulates the release of these hormones.

Ram, a resident of a remote Himalayan village, visited a doctor with complaints of tightness in the throat and a hoarse voice. The doctor advised a thyroid profile and a blood sugar test. Given below are the test results:

Test	Normal/Reference Range	Amount in Sample
TSH	0.4–5 mIU/L	6 mIU/L
T_3	4.6–12 μg/dL	1.76 μg/dL
T_4	80–180 ng/dL	21 ng/dL
Blood sugar (post-meal)	170–200 mg/dL	180 mg/dL

Based on this information, answer the following questions:

(A) What deficiency disorder might Ram be suffering from? Provide a justification.

(B) Mention two reasons why the doctor recommended a thyroid test.

(C) What role does iodine play in the production of thyroid hormones?

(D) Suggest a dietary recommendation for Ram to help him manage his condition if iodine deficiency is confirmed.

(A) Ram may be suffering from goitre due to iodine deficiency, indicated by throat tightness and hoarse voice.

(B) Thyroid test was advised to check:

1. Enlargement of the thyroid gland
2. Abnormal thyroid hormone levels

(C) Iodine is essential for the production of thyroxin.

(D) He should consume iodised salt and iodine-rich food.


Milte h Kal!